QUERIES:

1. CREATE DATABASES:
   mysql> CREATE DATABASE role_based_demo;
   Query OK, 1 row affected (0.01 sec)

   mysql> CREATE DATABASE analytics_db;
   Query OK, 1 row affected (0.01 sec)

2. CREATE TABLES IN BOTH DATABASES: mysql> -- In role_based_demo mysql> USE role_based_demo; Database changed mysql> CREATE TABLE users (-> id INT PRIMARY KEY,
   -> name VARCHAR(50),
   -> role VARCHAR(30)
   ->);
   Query OK, 0 rows affected (0.02 sec)

   mysql> mysql> -- In analytics_db
   mysql> USE analytics_db; Database changed mysql> CREATE TABLE reports (-> report_id INT PRIMARY KEY,
   -> report_name VARCHAR(100),
   -> created_on DATE
   ->);
   Query OK, 0 rows affected (0.02 sec)

3. CREATE ROLES:

```
mysql> CREATE ROLE 'system_admin';
Query OK, 0 rows affected (0.01 sec)

mysql> CREATE ROLE 'data_analyst_role';
Query OK, 0 rows affected (0.00 sec)

mysql> CREATE ROLE 'read_only_user';
Query OK, 0 rows affected (0.00 sec)
```

4. CREATE USERS:

```
mysql> CREATE USER 'sysadmin'@'localhost' IDENTIFIED BY 'password123';
Query OK, 0 rows affected (0.01 sec)

mysql> CREATE USER 'data_analyst'@'localhost' IDENTIFIED BY
'password123';
Query OK, 0 rows affected (0.01 sec)

mysql> CREATE USER 'readonly1'@'localhost' IDENTIFIED BY 'password123';
Query OK, 0 rows affected (0.01 sec)
```

5. GRANT PRIVILEGES TO ROLES:

```
mysql> -- System Admin: full control on both databases
mysql> GRANT ALL PRIVILEGES ON role_based_demo.* TO 'system_admin';
Query OK, 0 rows affected (0.01 sec)

mysql> GRANT ALL PRIVILEGES ON analytics_db.* TO 'system_admin';
Query OK, 0 rows affected (0.00 sec)

mysql> mysql> -- Data Analyst: read + update on users, read-only on
reports mysql> GRANT SELECT, UPDATE ON
role_based_demo.users TO
'data_analyst_role';
Query OK, 0 rows affected (0.00 sec)

mysql> GRANT SELECT ON analytics_db.reports TO 'data_analyst_role';
Query OK, 0 rows affected (0.00 sec)

mysql> mysql> -- Read-only user: only view
users table
mysql> GRANT SELECT ON role_based_demo.users TO 'read_only_user';
Query OK, 0 rows affected (0.00 sec)
```

6. ASSIGN ROLES TO USERS:

```
mysql> GRANT 'system_admin' TO 'sysadmin'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> GRANT 'data_analyst_role' TO 'data_analyst'@'localhost';
Query OK, 0 rows affected (0.00 sec)

mysql> GRANT 'read_only_user' TO 'readonly1'@'localhost';
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> SET DEFAULT ROLE ALL TO 'sysadmin'@'localhost';
Query OK, 0 rows affected (0.00 sec)

mysql> SET DEFAULT ROLE 'data_analyst_role' TO 'data_analyst'@'localhost';
Query OK, 0 rows affected (0.00 sec)

mysql> SET DEFAULT ROLE 'read_only_user' TO 'readonly1'@'localhost';
Query OK, 0 rows affected (0.00 sec)
```

7. ROW-LEVEL SECURITY SIMULATION (VIEWS):

```
mysql> CREATE VIEW analyst_view AS
    -> SELECT id, name FROM role_based_demo.users WHERE role = 'Analyst';
Query OK, 0 rows affected (0.01 sec)

mysql> mysql> GRANT SELECT ON
role_based_demo.analyst_view TO
'data_analyst_role';
ERROR 1146 (42S02): Table 'role_based_demo.analyst_view' doesn't exist
```

8. STORED PROCEDURE FOR CONTROLLED INSERT (ONLY ADMINS):

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE add_user(IN uid INT, IN uname
VARCHAR(50), IN urole VARCHAR(30))
    -> BEGIN
    ->     INSERT INTO role_based_demo.users VALUES (uid, uname, urole);
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ; mysql> mysql> GRANT EXECUTE ON PROCEDURE
role_based_demo.add_user TO
'system_admin';
ERROR 1305 (42000): PROCEDURE add_user does not exist
```

9. REVOKING PRIVILEGES (FOR SECURITY CHANGES):

```
mysql> REVOKE UPDATE ON role_based_demo.users FROM
```

'data_analyst_role';
Query OK, 0 rows affected (0.01 sec)

10. AUDITING – CREATE LOG TABLE AND TRIGGER:

```
mysql> CREATE TABLE audit_log (
    ->     action_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    ->     user_name VARCHAR(50),
    ->     action_performed VARCHAR(100)
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> mysql>
DELIMITER //
mysql> CREATE TRIGGER log_insert
   -> AFTER INSERT ON users
   -> FOR EACH ROW
   -> BEGIN
   ->     INSERT INTO audit_log (user_name, action_performed)
   ->     VALUES (CURRENT_USER(), CONCAT('Inserted user: ', NEW.name));
   -> END //
ERROR 1146 (42S02): Table 'analytics_db.users' doesn't exist mysql>
DELIMITER ;
```

11. SYSTEM ADMIN (SYSADMIN):

```
mysql> CALL add_user(1, 'John Doe', 'Admin'); -- works
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO analytics_db.reports VALUES (1, 'Monthly Report',
CURDATE()); -- works
Query OK, 1 row affected (0.00 sec)
```

12. DATA ANALYST (DATA_ANALYST):

```
mysql> SELECT * FROM analyst_view; -- sees only analysts
Empty set (0.00 sec)

mysql> UPDATE role_based_demo.users SET name = 'Updated Analyst' WHERE
id = 1; -- works (if not revoked)
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> DELETE FROM users WHERE id = 1; -- fails
ERROR 1146 (42S02): Table 'analytics_db.users' doesn't exist
```

13. READ ONLY USER (READONLY1):

```
mysql> SELECT * FROM role_based_demo.users; -- works +----+----------------
+-------+ | id | name           | role  |
+----+----------------+-------+
|  1 | Updated Analyst | Admin |
+----+----------------+-------+
1 row in set (0.00 sec)

mysql> INSERT INTO role_based_demo.users VALUES (2, 'Test', 'User'); -- fails
Query OK, 1 row affected (0.01 sec)
```